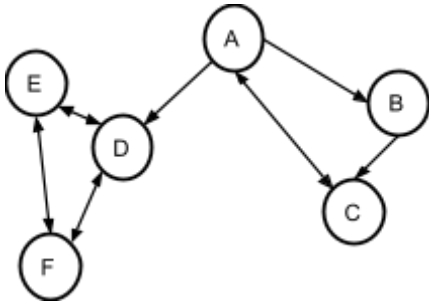
American Computer Science League

2023-2024 ● Contest 4: Short Problems ● Senior Division

1. Graph Theory

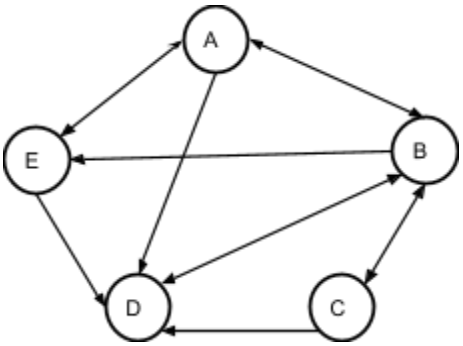
How many paths of length 4 are there from A in the directed graph below?



A. 13
B. 15
C. 18
D. 20
E. 21

2. Graph Theory

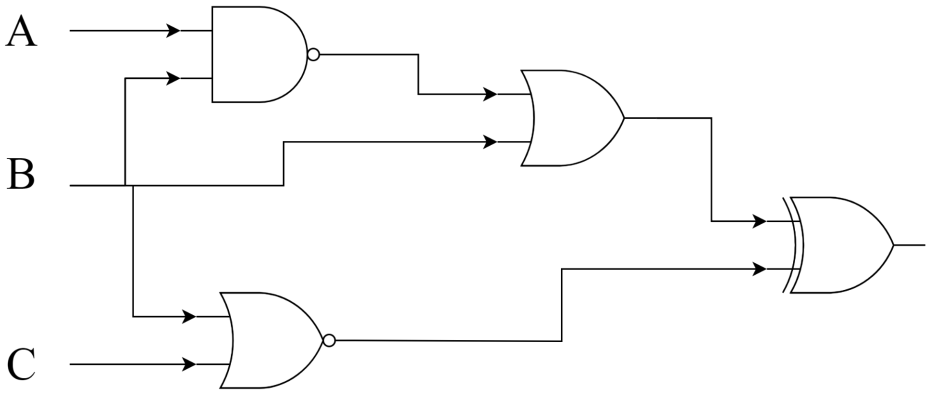
How many cycles are there in this directed graph?



A. 7
B. 8
C. 9
D. 10
E. 11

3. Digital Electronics

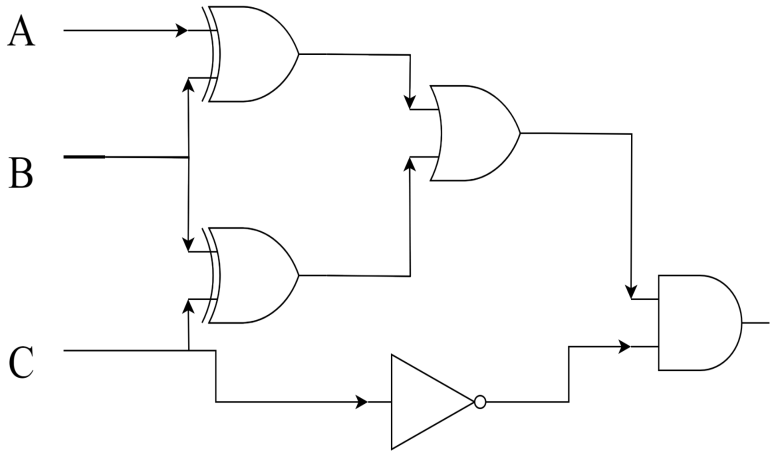
Simplify the Boolean expression represented by the following digital circuit:



A. $\overline{B} + C$
B. $\overline{B + C}$
C. $B + \overline{C}$
D. $\overline{BC}$
E. $\overline{B + C}$

4. Digital Electronics

How many ordered triples make the following digital circuit FALSE?



A. 2
B. 3
C. 4
D. 5
E. 6

5. Assembly Language

What is the output when the following program is executed?

```
T       DC      0
NUM     DC      28
        LOAD    =1
TOP     STORE   CNT
        LOAD    NUM
        DIV     =2
        SUB     CNT
        BL      STOP
        LOAD    NUM
        DIV     CNT
        MULT    CNT
        SUB     NUM
        BL      CONT
        LOAD    T
        ADD     CNT
        STORE   T
CONT    LOAD    CNT
        ADD     =1
        STORE   CNT
        BU      TOP
STOP    PRINT   T
        END
```

A. 10
B. 14
C. 28
D. 30
E. 56